Assignment: Concept Mapping Topics

Name: Abhidhya Sri Mailsamy Sumitha

Register Number: 192520051

Course: Cryptography and Network Security

Course code: CSA5116

Date: 17.07.2026

---

1. Security Attacks, Threats, Risks and Security Goals

Introduction

Information security aims to protect data, systems, and networks from unauthorized access, misuse, or damage. Security attacks exploit system vulnerabilities, creating threats that may lead to risks for individuals and organizations. The three fundamental security goals—**Confidentiality, Integrity, and Availability (CIA Triad)**—help protect information against these attacks.

---

a. How each attack leads to specific threats and risks

1. Unauthorized Access

Unauthorized access occurs when an attacker gains access to a system without permission.

Threats:

- Exposure of confidential information
- Data theft
- Identity theft

Risks:

- Financial losses
- Privacy violations
- Loss of customer trust

Security Goal:
Confidentiality

Real-world Example:
A hacker gains unauthorized access to a hospital database and steals patient medical records.

---

2. Malware Attack

Malware includes viruses, worms, ransomware, spyware, and Trojan horses designed to damage or disrupt computer systems.

Threats:

- File corruption
- Data modification
- System malfunction

Risks:

- Data loss
- Business interruption
- Financial damage

Security Goal:
Integrity

Real-world Example:
A ransomware attack encrypts a company's files, preventing employees from accessing important business documents.

---

3. Phishing Attack

Phishing is a social engineering attack where attackers trick users into revealing sensitive information.

Threats:

- Password theft
- Credential compromise
- Identity theft

Risks:

- Unauthorized transactions
- Financial fraud
- Data breaches

Security Goal:
Confidentiality

Real-world Example:
An employee receives a fake banking email and unknowingly enters login credentials into a fraudulent website.

---

4. Denial-of-Service (DoS) Attack

A Denial-of-Service attack floods a server or network with excessive requests, making it unavailable to legitimate users.

Threats:

- Network congestion
- Service interruption

Risks:

- Business downtime
- Revenue loss
- Poor customer satisfaction

Security Goal:
Availability

Real-world Example:
An online shopping website becomes inaccessible during a festive sale due to a Distributed Denial-of-Service (DDoS) attack.

---

b. How Security Goals Help Mitigate Attacks

Confidentiality

Confidentiality ensures that only authorized users can access sensitive information. It is maintained through encryption, authentication, passwords, and access control mechanisms.

Example:
Banks use encrypted HTTPS connections to protect customers' financial information.

Integrity

Integrity ensures that information remains accurate and unaltered during storage and transmission. Techniques such as hash functions, checksums, and digital signatures are used to maintain integrity.

Example:
Software companies publish SHA-256 hash values so users can verify downloaded software has not been modified.

Availability

Availability ensures that systems and data remain accessible whenever needed. It is maintained using backup servers, firewalls, redundancy, and disaster recovery plans.

Example:
Cloud providers maintain multiple data centers so services remain available even if one server fails.

Conclusion

Security attacks create threats that eventually lead to organizational risks. Implementing the CIA Triad helps organizations reduce these risks and maintain secure communication and reliable information systems.

---

2. Comparison of Substitution Techniques

Introduction

Substitution ciphers encrypt messages by replacing plaintext characters with different ciphertext characters. Different substitution techniques provide varying levels of security depending on key management and encryption complexity.

Caesar Cipher

The Caesar Cipher is one of the simplest substitution techniques. Each letter is shifted by a fixed number of positions in the alphabet.

Key Usage:
Uses a single numerical shift value.

Complexity:
Very low.

Security Strength:
Weak because there are only 25 possible keys, making brute-force attacks easy.

Monoalphabetic Cipher

The Monoalphabetic Cipher replaces each plaintext letter with another fixed letter according to a substitution alphabet.

Key Usage:
Uses one substitution alphabet.

Complexity:
Moderate.

Security Strength:
More secure than the Caesar Cipher but vulnerable to frequency analysis.

Vigenère Cipher

The Vigenère Cipher uses a repeating keyword to determine different shift values for different letters.

Key Usage:
Uses a repeating keyword.

Complexity:
High.

Security Strength:
Provides better security because identical plaintext letters can produce different ciphertext letters.

Hill Cipher

The Hill Cipher uses matrix multiplication to encrypt groups of letters instead of individual characters.

Key Usage:
Uses an invertible matrix as the encryption key.

Complexity:
Very high.

Security Strength:
Stronger than other classical substitution ciphers because it encrypts blocks of letters and provides greater diffusion.

Analysis

Among the four techniques, the **Hill Cipher** is the most secure because it encrypts multiple characters simultaneously using matrix operations. This makes frequency analysis much more difficult compared to Caesar, Monoalphabetic, and Vigenère ciphers. However, modern encryption algorithms such as AES are significantly more secure than any classical substitution technique.

Conclusion

As encryption complexity increases, security also improves. The Hill Cipher offers the highest level of security among classical substitution ciphers because of its mathematical foundation and block-based encryption.

---

3. Classical Encryption Schemes: Substitution and Transposition Techniques

Introduction

Classical encryption methods are broadly classified into substitution techniques and transposition techniques. Both aim to protect information but use different approaches to transform plaintext into ciphertext.

---

a. Differences and Similarities

Substitution Techniques

Substitution techniques replace each plaintext character with another character according to a specific rule.

Examples include Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher, and Hill Cipher.

Transposition Techniques

Transposition techniques rearrange the positions of characters without changing the actual characters.

Examples include Rail Fence Cipher and Columnar Transposition Cipher.

Similarities

- Both convert plaintext into ciphertext.
- Both require encryption and decryption procedures.
- Both are classical cryptographic techniques.
- Both improve confidentiality compared to plain text communication.

---

i. Effect on Data Structure and Security Level

Substitution techniques change the identity of characters while preserving their positions. This alters the appearance of the message and provides moderate security.

Transposition techniques preserve the original characters but change their order. Although the message appears scrambled, the original letter frequencies remain unchanged, resulting in lower security.

---

ii. Which Technique is More Vulnerable?

Transposition techniques are generally more vulnerable because they only rearrange existing letters. Since character frequencies remain unchanged, attackers can use frequency analysis and brute-force attacks to recover the original message.

Substitution techniques provide comparatively better security because character values are altered. However, simple substitution methods are still vulnerable to frequency analysis.

Conclusion

Although substitution techniques generally provide better protection than transposition techniques, both are considered insecure by modern standards and should only be used for educational purposes.

---

4. Steganography Concepts: Data Hiding, Watermarking and Survivability

Introduction

Steganography is the science of hiding secret information inside digital media such as images, audio, or videos. Unlike cryptography, which hides the content of a message, steganography hides the existence of the message itself.

---

a. Relationship Between the Concepts

Data Hiding involves embedding confidential information inside another digital file without noticeably changing its appearance.

Watermarking embeds ownership or copyright information into digital media to prove authenticity and prevent unauthorized copying.

Survivability refers to the ability of hidden information to remain intact even after compression, resizing, filtering, or transmission.

These three concepts work together to improve the security and reliability of hidden communications.

---

b. Contribution to Secure Communication

Data hiding ensures that confidential information remains invisible to unauthorized users.

Watermarking protects intellectual property by proving ownership and authenticity.

Survivability ensures that hidden information can still be recovered even after normal processing of the digital media.

Together, these techniques provide multiple layers of protection for secure communication.

---

c. Real-world Applications

Data Hiding
Military organizations hide confidential messages inside satellite images during secure communication.

Watermarking
Photographers and movie studios embed digital watermarks into images and videos to protect copyrights.

Survivability
Hospitals embed patient information inside medical images so that the information remains available even after image compression and transmission.

Conclusion

Steganography complements cryptography by concealing the existence of confidential information. Data hiding, watermarking, and survivability together improve the confidentiality, authenticity, and reliability of digital communication.

---

5. Number Theory Concepts Used in Cryptography

Introduction

Number theory forms the mathematical foundation of modern cryptography. Public-key cryptographic algorithms such as RSA depend heavily on modular arithmetic and related mathematical concepts.

---

a. Mathematical Connections

The **Extended Euclidean Algorithm** calculates the Greatest Common Divisor (GCD) of two integers and is used to determine modular inverses.

The **Modular Inverse** allows division in modular arithmetic and is essential for RSA key generation.

Euler's Phi Function calculates the number of integers less than a given number that are relatively prime to it.

Euler's Theorem states that if two integers are relatively prime, modular exponentiation follows a predictable mathematical relationship based on Euler's Phi Function.

Fermat's Little Theorem is a special case of Euler's Theorem when the modulus is a prime number.

Together, these concepts form the mathematical framework used in public-key cryptography.

---

b. Role in Encryption and Decryption

The Extended Euclidean Algorithm computes the modular inverse required to generate RSA private keys.

The Modular Inverse is used to calculate the private exponent used during decryption.

Euler's Phi Function is required during RSA key generation and determines the relationship between the public and private keys.

Euler's Theorem provides the mathematical proof that RSA encryption and decryption correctly recover the original plaintext.

Fermat's Little Theorem is widely used for modular exponentiation and efficient primality testing.

---

c. Most Critical Concept for Public-Key Cryptography

Among these concepts, **Euler's Phi Function** is considered the most critical because it is directly involved in RSA key generation. It enables the calculation of the private key through modular inverses and ensures that encryption and decryption work correctly.

Without Euler's Phi Function, the RSA algorithm cannot generate valid public and private key pairs. Although the Extended Euclidean Algorithm and Modular Inverse are also essential, they depend on Euler's Phi Function for their application in RSA.

Conclusion

Number theory provides the mathematical principles required for secure public-key cryptography. Concepts such as the Extended Euclidean Algorithm, Modular Inverse, Euler's Phi Function, Euler's Theorem, and Fermat's Little Theorem work together to enable secure encryption, decryption, digital signatures, and authentication in modern communication systems.